

GRADE 10 ICT

Logic Gates with Boolean Functions

Module 4.1 – Introduction to Digital Logic and Basic
Logic Concepts

MODULE 4.1

Introduction to Digital Logic and Basic Logic Concepts



What is a Signal?

Information used to communicate a condition or message.

-  **Green light triggers train movement.**
-  **Red light stops the train.**
-  **Alarm sounds require immediate attention.**

Everyday Signals

Signals help systems identify conditions and make decisions.



Safe Access

Two keys open the safe.



Door Alert

Open door triggers a warning.



Safety Belt

Unbuckled belt sounds alarm.

Digital Signals

Computers represent information using two specific states.

0

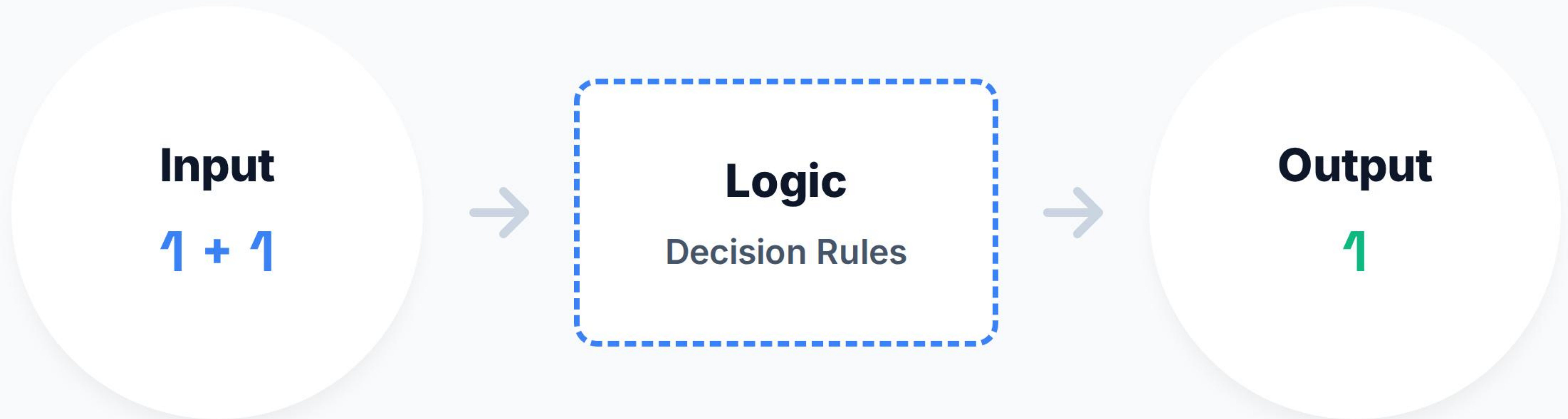
OFF / False

1

ON / True

What is Digital Logic?

A system of rules used to process binary states and produce an output.

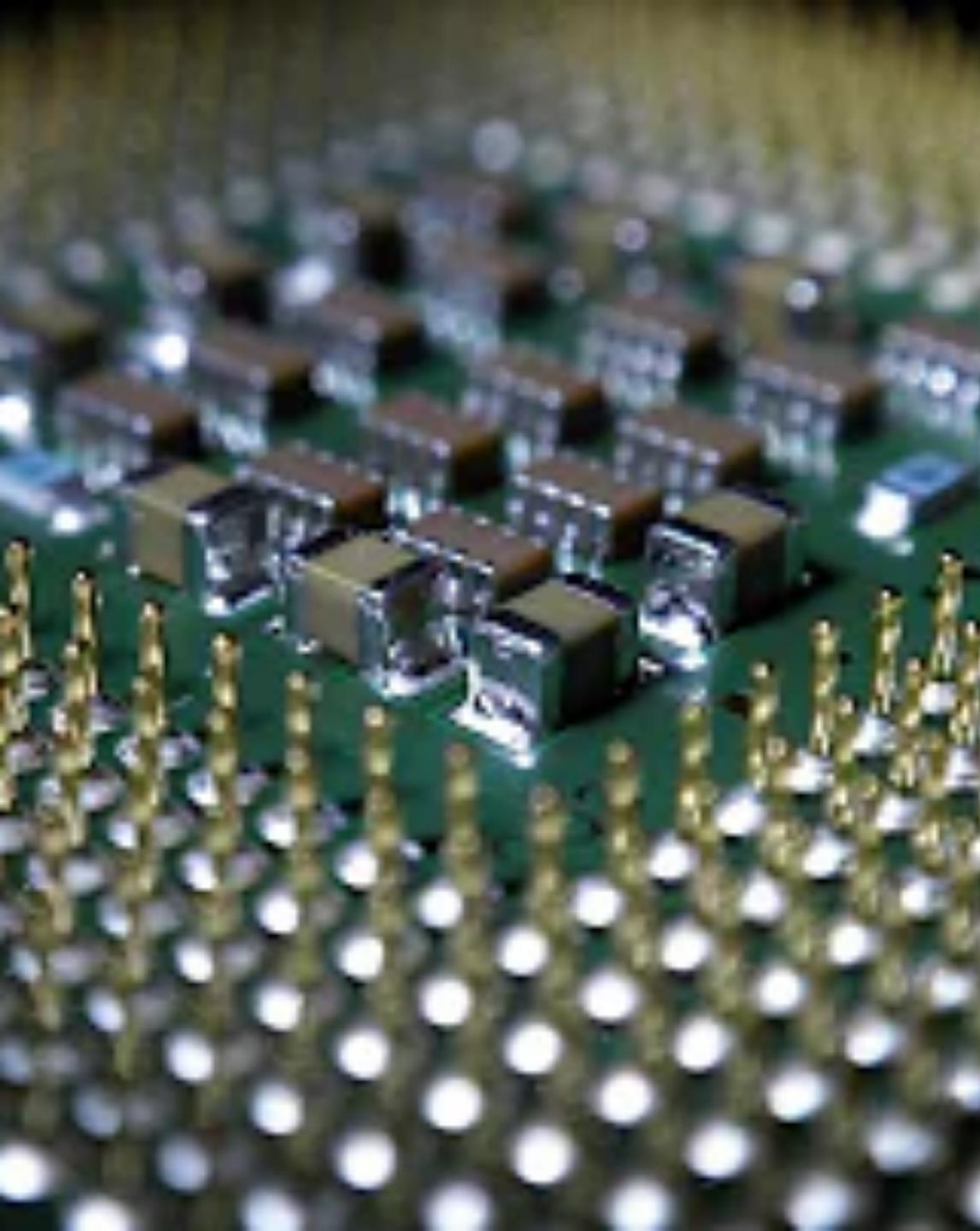


The Logic Circuit

An electronic circuit utilizing binary values to execute decisions.

-  **Receives Inputs**
-  **Processes Data**
-  **Produces Binary Output**





The Logic Gate

The basic building block of digital circuits.

A logic gate is a basic digital circuit that performs a specific logical operation.

Input → Logical Operation → Output

Classifications of Logic Gates

1. Basic Logic Gates

AND

OR

NOT

2. Combinational Gates

NOR

NAND

Created by combining basic gates.

Inputs and Outputs

The final output Q depends entirely on the input values of A and B.



Binary Logic in Practice



Switch State

Open

0

Closed

1



Bulb State

OFF

0

ON

1

Quick Activity

A door opens ONLY when two specific keys are available simultaneously.

Key A: 0	Key B: 0	→	Stays Closed
Key A: 1	Key B: 0	→	Stays Closed
Key A: 0	Key B: 1	→	Stays Closed
Key A: 1	Key B: 1	→	Door Opens

 **AND Gate**